

OPERATIONAL RISK MANAGEMENT AND REGULATION

(DEFINITION, TYPES, BIA, STA, AMA, SMA, MODELLING OPERATIONAL LOSSES,
REGULATORY REQUIREMENTS, CRISIS RISK)

Sources:

- *John C. Hull - Risk Management and Financial Institutions-Wiley (2015) Chapter 23.*
- *Lecture slides*

It is much more difficult to quantify and manage operational risk than credit or market risk. Financial institutions make a conscious decision to take a certain amount of credit and market risk, and there are many traded instruments that can be used to reduce these risks. Operational risk, by contrast, is a necessary part of doing business. An important part of operational risk management is *identifying the types of risks* that are being taken and which should be insured against. There is always a danger that a huge loss will be incurred from taking an operational risk that ex ante was not even recognized as a risk.

It is tempting to consider operational risk as a residual risk and define it as any risk faced by a financial institution that is not market risk or credit risk. Most people agree that this definition of operational risk is too broad. It includes the risks associated with entering new markets, developing new products, economic factors, and so on. Another possible definition is that operational risk, as its name implies, is the risk arising from operations. This includes the risk of mistakes in processing transactions, making payments, and so on. This definition of risk is too narrow.

Definition: *Operational risk is the risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events.* This definition includes legal risk, but not strategic and reputational risk. We can distinguish between *internal risks* and *external risks*. **Internal risks** are those over which the company has control. The company chooses whom it employs, what computer systems it develops, what controls are in place, and so on. Operational risk then includes more than just the risk arising from operations. It includes risks arising from inadequate controls such as the risk of employee fraud. Bank regulators favor including more than just internal risks in their definition of operational risk. They include the impact of **external events** such as natural disasters (for example, a fire or an earthquake), political and regulatory / legal risk (for example, being prevented from operating in a foreign country by that country's

government; non-compliance to legal regulations), security breaches, and so on. The greatest operational risks based on previous year surveys are: *data compromise, problems with the IT system, theft and fraud*.

Some operational risks *interact with credit and market risk*. For example, when mistakes are made in a loan's documentation, it is usually the case that losses result if and only if the counterparty defaults. Or, when a trader exceeds limits and misreports positions, losses result if and only if the market moves against the trader.

Categories

The Basel Committee on Banking Supervision (BCBS) has identified seven categories of operational risk:

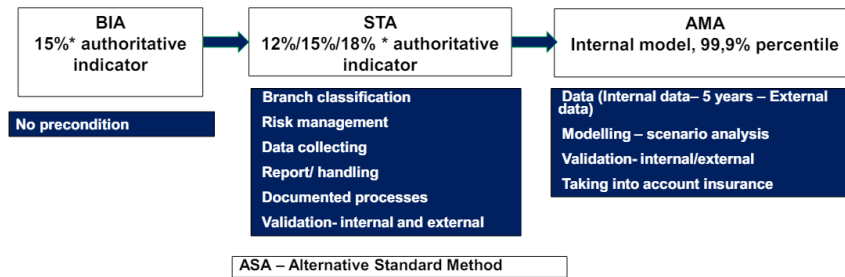
1. **Internal fraud**: deliberate act of fraud in which at least one party is an employee of the organization. Examples include falsification of financial statements, employee theft, and insider trading on an employee's own account.
2. **External fraud**: deliberate act of fraud by a third party (customers or external entities). Examples include robbery or cyber-attacks.
3. **Employment practices and workplace safety**: inappropriate or unsafe treatment of employees or breaches of employment laws. Examples include workers compensation claims, violation of employee health and safety rules, organized labor activities, discrimination claims, harassment claims, and general liability (for example, a customer slipping and falling at a branch office).
4. **Clients, products, and business practices**: unintentional damage done to customers by the organization's products. Examples are misleading customer information, or physical damage.
5. **Damage to physical assets**: Loss or damage to physical assets from natural disasters or other events. Examples include terrorism, vandalism, earthquakes, fires, and floods.
6. **Business disruption and system failures**: disruptions to operations or IT systems (IT, telecommunication). Examples include hardware and software failures, telecommunication problems, and utility outages.
7. **Execution, delivery, and process management**: errors or inefficiencies in transaction processing, data management, or process design. Examples include data entry errors, collateral management failures, incomplete legal documentation, missed deadlines for regulatory filings, nonclient counterparty mis-performance, and vendor disputes.

Capital calculation methods

Capital requirements for operational risk were introduced in *Basel II*. Banks have 3 alternatives to calculate operational risk regulatory capital under Basel II.

There are some conditions that a bank must satisfy in order to use the standardized approach or the AMA approach.

Capital calculation methods



From 2025 SMA method: Gross income based, corrected by losses

- **BIA (Basic Indicator Approach)**

It is the simplest approach. Under this approach, operational risk capital is set equal to *15% of average annual gross income* over the previous 3 years. Gross income is defined as net interest income plus non-interest income.

- **STA (Standardized Approach)**

It is a slightly more complicated approach. In this, a bank's activities are divided into *eight business lines* (ex. corporate finance, retail banking, commercial banking etc.). The *average annual gross income* over the last 3 years for each business line is multiplied by a "*beta factor*" (12%/15%/18%) corresponding to that business line and the *results are summed* to determine the total capital.

- **AMA (Advanced Measurement Approach)**

It is the most complex approach. In this approach, the regulatory capital requirement is calculated based on banks' internal models. It is based on a *1-year 99.9% VaR*. The objective of banks using the AMA approach for operational risk is similar to their objectives when they attempt to measure credit risk: produce a *probability distribution of losses*. Assuming that they can convince regulators that their *expected operational risk cost* is incorporated into their pricing of products, *capital is assigned to cover unexpected costs*. The unexpected loss is the difference between the **1-year 99.9% VaR** and the one-year expected loss.

It was phased-out due to its complexity and resource intensity.

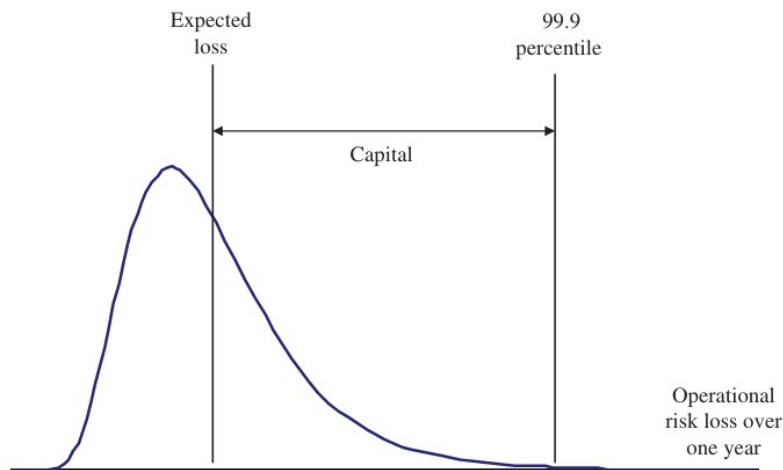


FIGURE 23.1 Calculation of Capital for Operational Risk

- **SMA**

The Standardized Measurement Approach (SMA) for determining operational risk capital requirements was developed as part of the Basel III and is intended to *replace all previous approaches*. It aimed to simplify and standardize the calculation of operational risk capital.

- **Methodology:** It combines a *business indicator component (BIC)*, which is based on financial statement items that reflect the bank's size and income, with a *loss multiplier (LM)*. The LM adjusts the capital requirement based on the bank's historical operational loss experience.

Calculation of SMA

- **Business Indicator Component (BIC):** The BIC is the sum of the banks *business indicators* (Interest, Lease, and Dividend Component; Services Component; Financial Component) which reflect the size of the bank. Then based on their size, banks are grouped into buckets, larger banks ending up with larger BIC (thus larger capital requirement).
- **Loss Component (LC):** This component involves calculating a *loss multiplier* based on the bank's *10-year historical operational loss experience*. The loss multiplier adjusts the BIC, increasing the capital requirement for banks with poor operational loss histories and reducing it for those with better loss experiences.

For banks with larger than 1 billion BI the SMA capital is:

$$\text{SMA Capital} = 110 + (\text{BI Component} - 110) * \text{Internal Loss Multiplier}$$

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Modelling Operational Losses

Banks calculate two distributions: *loss frequency distribution* and *loss severity distribution*. The *loss frequency distribution* is the distribution of the number of losses observed for one year. The *loss severity distribution* is the distribution of the size of a loss, given that a loss occurs.

For loss frequency *Poisson distribution* is used (mean and variance of frequency of losses are equal), assuming that events occur randomly and independently. The probability of n losses in T years is:

$$e^{-\lambda T} \frac{(\lambda T)^n}{n!}$$

where λ is the average number of losses per year.

For loss severity *lognormal distribution* is used (with mean and sd of the logarithm of the loss), skewed to capture rare but extreme losses.

The *loss distribution* can be determined with Monte Carlo simulation by *combining the loss frequency and the loss severity distributions*.

1. We sample from the frequency distribution to determine the number of loss events ($= n$) in one year.
2. We sample n times from the loss severity distribution to determine the loss experienced for each loss event ($L_1, L_2, \dots, L_n$).
3. We determine the total loss experienced ($= L_1 + L_2 + \dots + L_n$).

When many simulation trials are used, we obtain a total loss distribution for losses of the type being considered. The 99.9 percentile of the distribution can then be determined.

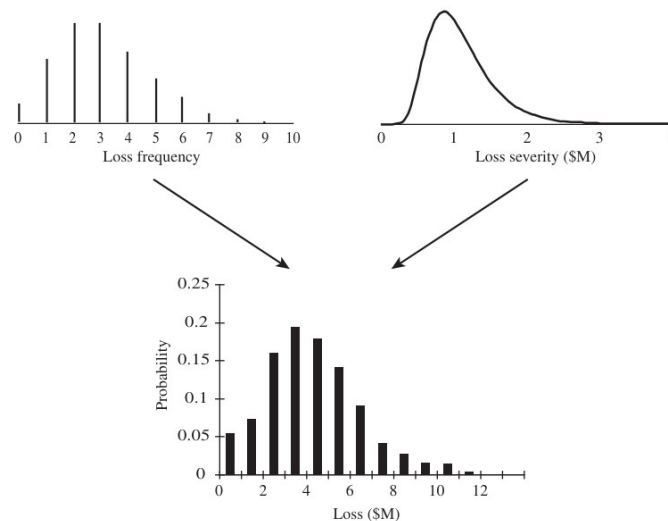


FIGURE 23.2 Calculation of Loss Distribution from Loss Frequency and Loss Severity

Modelling operational losses involves three important elements: *internal data*, *external data*, *scenario analysis*, and *business environment and internal control factors*.

- Internal Data

Unfortunately, there is usually relatively little historical data available within a bank to estimate *loss severity* and *loss frequency distributions* for particular types of losses. Many banks have not kept records in the past of operational risk losses, such as has been done for credit risk (they are doing so now).

There are two types of operational risk losses: *high-frequency low-severity losses* (HFLSLs) and *low-frequency high-severity losses* (LFHSLs). An example of the first is credit card fraud losses. An example of the second is rogue trader losses.

A bank should *focus its attention on LFHSLs*. These are what *create the tail of the loss distribution*. Another reason for focusing on LFHSLs is that *HFLSLs are often taken into account in the pricing of products*.

By definition, LFHSLs occur infrequently. Even if good records have been kept, internal data are liable to be inadequate and must be supplemented with external data and scenario analysis.

- External Data

External data is used to determine loss severity. There are two sources of external data. The first is *data consortia*, which are companies dealing with data sharing between banks. (The insurance industry has had mechanisms for sharing loss data for many years and banks are now doing this as well.) The second is *data vendors*, who are collecting publicly available data in a systematic way. The loss data available from data vendors is *data taken from public sources* such as newspapers and trade journals. Data from vendors cannot be used in the same way as internal data or data obtained through sharing arrangements because they are *subject to biases*. For example, only large losses are publicly reported, and the larger the loss, the more likely it is to be reported.

External data increases the amount of data available to a bank for estimating potential losses. It also has the advantage that it can lead to a *consideration of types of losses* that have never been incurred by the bank, but *which have been incurred by other banks*. Both internal and external historical data must be adjusted for inflation. In addition, a scale adjustment should be made to external data. If a bank with a revenue of \$10 billion reports a loss of \$8 million, how should the loss be scaled for a bank with a revenue of \$5 billion? A natural assumption is that a similar loss for a bank with a revenue of \$5 billion would be \$4 million. But this estimate is probably too

small. For example, research by Shih et al. (2000) suggests that the effect of firm size on the size of a loss experienced is non-linear. Their estimate is:

$$\text{Estimated Loss for Bank A} = \text{Observed Loss for Bank B} \times \left(\frac{\text{Bank A Revenue}}{\text{Bank B Revenue}} \right)^{\alpha}$$

where $\alpha = 0.23$. This means that in our example the bank with a revenue of \$5 billion would experience a loss of $8 \times 0.50^{0.23} = \$6.82$ million. After the appropriate scale adjustment, data obtained through sharing arrangements with other banks can be merged with the bank's own data to obtain a larger sample for determining the loss severity distribution.

- **Scenario Analysis**

Scenario analysis has become a key tool for the assessment of operational risk under the AMA. The aim of scenario analysis is to *generate scenarios covering the full range of possible LFHSLs*. Scenarios might come from the bank's own experience, might be based on the experience of other banks or might be generated by the risk management group.

An operational risk committee consisting of members of the risk management group and senior management should be asked to *estimate loss severity and loss frequency parameters* for the scenarios. Data from other banks may be useful for estimating the loss severity parameters. The loss frequency parameters should reflect the controls in place at the bank and the type of business it is doing. They should reflect the views of the members of the operational risk committee.

Whether scenario analysis or internal/external data approaches are used, distributions for particular loss types have to be combined to produce a total operational risk loss distribution. The correlations assumed for the losses from different operational risk categories can make a big difference to the 1-year 99.9% VaR that is calculated, and therefore to the AMA capital. It is *often argued that operational risk losses are largely uncorrelated* with each other and there is some empirical support for this view. If the zero-correlation assumption is made, Monte Carlo simulation can be used in a straightforward way to sample from the distribution of losses for each scenario to obtain a total distribution of risk losses.

- **Business Environment and Internal Control Factors**

Business environment and internal control factors (BEICFs) should be taken into account when loss severity and loss frequency are estimated. These include the complexity of the business line, the technology used, the pace of change, the level of supervision, staff turnover rates, and so on.

Crisis risk

Crisis is an event or action that becomes public, potentially endangering trust in the company or damaging its reputation, that cannot be handled as part of the company's normal operation.

It is a process, rather than an event; it is a source of uncertainty threatening the company and its stakeholders.

Crisis management has 3 steps:

- Crisis prevention, preparation
- Tasks to be performed during a crisis
- Post-crisis assessment

There are 2 crisis management approaches:

Internal approach

- **Pre-crisis:** organizational readiness/preparedness
 - How prepared the company is for a crisis (policies and frameworks for crisis response)
- **Crisis:** leadership at crisis
 - Leadership quality in handling the crisis
 - Internal communication (clear and consistent communication with employees and management)
- **Post-crisis:** organizational learning
 - Individual development, motivation, alternative solution search (investigating and addressing the underlying causes of the crisis to prevent recurrence)
 - Learning from others' mistakes

External approach

- **Pre-crisis:** relationship with stakeholders
 - Realistic expectations and positive relationship
- **Crisis:** stakeholders' expectations
 - Ensuring transparent communication with external stakeholders
 - Strategies in response to the crisis, time factor
 - How responsible is the company? → reputation
 - How does it influence the stakeholders?
- **Post-crisis:** social judgement

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- The social assessment of the responses to the crisis